Keyframe Extraction assisted Crime Detection

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***Abstract*—This paper investigates the impact of keyframe extraction algorithms on the accuracy of the action detection model. The work aims to improve the efficiency of crime detection systems by enabling real-time inference and accurate predictions. The performance of various state-of-the-art models with different keyframe extraction algorithms is evaluated. The results highlight the significant effect of the algorithm on model accuracy, underscoring the importance of selecting an appropriate keyframe extraction algorithm for effective action detection. Additionally, a parameter-efficient fine-tuning approach is proposed that can reduce the required training data while improving the model's performance, which could further enhance the system's efficiency. The models are evaluated on a publicly available dataset to ensure comparability, and the study thoroughly evaluates the proposed approach.**

***Index Terms*—Agent Profiling, Slow Fast, Frame Extraction, Crime Detection**

I. Introduction

The development of efficient crime detection systems [6] has gained significant attention in recent years, with the aim of improving public safety and reducing response time. One crucial aspect of these systems is accurate action detection, which involves recognizing and classifying human actions in real-time video streams [2]. The accuracy of action detection models heavily relies on the extraction of keyframes, which represent the most informative frames for action recognition. This paper delves into the impact of keyframe extraction algorithms on the accuracy of action detection models, with the ultimate goal of enhancing the efficiency of crime detection systems.

The problem of accurately detecting human actions in real-time video streams poses several challenges [4]. First and foremost, the sheer volume of visual data makes it computationally intensive to process each frame individually. Keyframe extraction algorithms offer a potential solution by selecting a subset of frames that capture the essential information needed for action recognition. However, the effectiveness of these algorithms in preserving the accuracy of action detection models remains an open question. Furthermore, the selection of an appropriate keyframe extraction algorithm plays a critical role in achieving real-time inference and accurate predictions in crime detection systems.

The state of the art in keyframe extraction algorithms for action detection has witnessed significant advancements. Several approaches have been proposed, including heuristic-based methods, saliency-based methods, and deep learning-based methods. These algorithms aim to identify frames that encapsulate the essential temporal and spatial characteristics of human actions. However, there is a lack of comprehensive studies that evaluate the impact of these algorithms on the accuracy of action detection models. The strengths and weaknesses of different keyframe extraction algorithms need to be understood, and the most suitable approach that can maximize the efficiency of crime detection systems must be selected. To address the aforementioned challenges, a systematic investigation of the impact of keyframe extraction algorithms on the accuracy of action detection models is presented in this paper. The performance of various state-of-the-art models using different keyframe extraction algorithms is evaluated. The experimental results reveal the substantial influence of the algorithm on the accuracy of the action detection models. These findings emphasize the importance of selecting an appropriate keyframe extraction algorithm to ensure effective action detection in crime detection systems. The compute-efficient aspect of the model allows real-time usage of the same and is tested by building a web application on top of the model for real-time inference. The key contributions of the paper are:

- Evaluate the performance of state-of-the-art action recognition models on top of key frame extraction algorithms and benchmark the performance on UCF Crime dataset.
- Propose a parameter-efficient fine-tuning approach for action recognition models that demands 90% less training data and results in 1-2% performance gains.

In addition to evaluating the existing algorithms, a parameter-efficient fine-tuning approach is proposed that can improve the performance of action detection models while reducing the required training data. [3] This approach aims to enhance the efficiency of the system by achieving higher accuracy with a smaller training dataset. By leveraging the strengths of the selected keyframe extraction algorithm, the proposed approach offers a promising solution to improve the effectiveness of action detection models in real-time crime detection scenarios.

The remainder of this paper is organized as follows. Section 2 provides an overview of related work in the field of keyframe extraction algorithms for action detection. Section 3 details the methodology used in the experimental setup and introduces the publicly available dataset used for evaluation. Section 4 presents the experimental results and discusses the findings. Section 5 introduces the proposed parameter-efficient fine-tuning approach. Finally, Section 6 concludes the paper by summarizing the contributions and outlining potential future research directions.

II. Literature Survey

We divide this section in two parts based on the major aspects of our research. The first part II-A focuses on previous work done in the action detection domain whereas the second part II-C highlights previous work done in the keyframe extraction domain.

A. Action Detection

Several deep learning models have been proposed for action detection, each with its unique architecture and strengths. This literature survey reviews some of the most prominent action detection models in the literature and discusses their key features, strengths, and limitations. We will then focus on three specific models, namely Temporal Interlacing Network (TIN), SlowFast, and UniformerV2, and explain why we have chosen these models for our research project. We will describe the specific features of each model and discuss how they differ from each other, highlighting their potential advantages and limitations for action detection in videos.

SlowFast [10] is a deep neural network for action detection in videos that leverages a two-stream architecture. It combines a "slow" stream, capturing long-term temporal information and spatial context, with a "fast" stream, capturing short-term temporal information and motion cues. By fusing the outputs of these streams at multiple levels, SlowFast achieves a comprehensive understanding of video content. Strengths of SlowFast include its ability to handle both long-term and short-term temporal information, its efficient architecture suitable for real-time applications, and its potential for large-scale video analysis. However, limitations exist, such as reliance on large labeled datasets for training and challenges in handling complex scenes. Biases can also arise from dataset choices and design decisions. Future research should address these weaknesses and biases for improved action detection in real-world crime scenarios.

UniFormerV2, [11] a family of video networks, revolutionizes video understanding by enhancing pre-trained Vision Transformers (ViTs) with efficient UniFormer designs. By unifying convolution and self-attention as a relation aggregator in the transformer format, UniFormer mitigates local video redundancy in ViTs. UniFormerV2 inherits the concise style of the UniFormer block and introduces novel local and global relation aggregators, integrating the advantages of both ViTs and UniFormer. These aggregators enable UniFormerV2 to capture short-term temporal information, motion cues, long-term temporal information, and spatial context. UniFormerV2, with its integration of Vision Transformers (ViTs) and UniFormer designs, offers a solution to the problem of local video redundancy. By combining convolution and self-attention as a relation aggregator, UniFormerV2 captures both short-term and long-term temporal information in videos, enabling comprehensive understanding. Its architecture distinguishes it from similar models, and its potential applications include action recognition, object detection, video segmentation, surveillance systems, content moderation, anomaly detection, video search, recommendation systems, and video summarization. However, challenges remain in effectively capturing complex spatio-temporal relationships, and reliance on pre-trained ViTs may pose computational and data requirements.

TIN(Temporal Interlacing Network) [12] is a deep neural network that employs temporal interlacing to capture more temporal information and detect fast-changing actions. With inception-style convolutional layers and a spatial-temporal attention mechanism, TIN effectively extracts spatial and temporal features. While it achieves state-of-the-art performance in action detection, limitations include a potential loss of subtle temporal details due to lower frame rates and increased computational complexity. TIN's unique architecture sets it apart from similar models, and its applications range from action recognition to video surveillance, where the detection of fast-changing actions is vital.

As for the differences in the models, TIN uses a simpler single-stream architecture, while Uniformer V2 and SlowFast both use more complex two-stage architectures that combine proposal generation and action classification. Additionally, Uniformer V2 and SlowFast both use multi-stream architectures that process different temporal resolutions of the input data, while TIN only processes frames at a single temporal resolution. Finally, SlowFast uses a "top-down" network design to reduce the computational cost, which is not used in TIN or Uniformer V2. Using these three models with different architectures allows us to evaluate the performance of different keyframe extraction methods for action detection in videos. By testing how these models perform when trained on different types of keyframes, we can compare the effectiveness of different keyframe extraction techniques and determine which methods are most suitable for action detection in different scenarios.

B. Keyframe extraction

VSUMM, SIFT, and the histogram-based keyframe extraction algorithm are three prominent approaches for video keyframe extraction, each with its own strengths and limitations. VSUMM [13] (Video Summarization Using Multi-Clustering) excels at capturing relevance by utilizing clustering techniques to group similar frames and selecting representative frames, providing a comprehensive summary. However, VSUMM is sensitive to clustering and heuristic choices, impacting the quality of the extracted keyframes. SIFT (Scale-Invariant Feature Transform) [16] detects distinctive features invariant to scale, rotation, and translation, making it robust for identifying important moments. Yet, it may miss keyframes lacking distinctive features. On the other hand, the histogram-based [6] method focuses on significant color distribution changes, offering simplicity and adaptability. However, it may overlook other relevant aspects of video content. These differences highlight trade-offs between relevance, distinctive features, and color changes, making each algorithm suitable for specific use cases in video analysis and summarization. VSUMM provides a comprehensive summary but relies on appropriate clustering and heuristic selection. SIFT is robust in capturing distinctive moments but may overlook non-distinctive keyframes, while the histogram-based approach captures color changes but may miss other significant content. Understanding these pros and cons aids in selecting the most suitable algorithm based on the specific requirements of the application.

C. Dataset

The UCF-Crime dataset constitutes a substantial archive of 128 hours of video material, encompassing 1900 unaltered, real-world surveillance recordings, highlighting 13 prevalent real-life irregularities. These aberrations encompass behaviors such as Abuse, Arrest, Arson, Assault, Road Accident, Burglary, Explosion, Fighting, Robbery, Shooting, Stealing, Shoplifting, and Vandalism. The selection of these anomalies is predicated upon their potential to constitute noteworthy threats to public safety. Researchers have extensively harnessed this dataset to construct and assess algorithms tailored to crime detection, anomaly identification, and activity recognition tasks. The UCF Crime Dataset's merits are rooted in its lifelike and demanding scenarios, facilitating the creation of sturdy and precise video analysis models. Nevertheless, it is imperative to recognize certain limitations, including the dataset's relatively finite size and the potential for biases in the assortment of represented crimes.

III. Research Contribution

This paper addresses several research inquiries pertaining to the training of large-scale vision models on extensive video datasets with computational efficiency while ensuring per-class performance. The study investigates the consequences of augmenting and selecting frames before their input into a vision model for action recognition. Additionally, the paper explores the identification of the most optimal architecture for detecting crime actions in lengthy videos, emphasizing the minimization of computational resource requirements during both the training and inference stages.

The present research study significantly contributes to the field of action detection and keyframe extraction by undertaking a thorough investigation into the efficacy of combining three prominent action detection models, namely SlowFast, UniformerV2, and TIN, with three keyframe extraction algorithms, namely VSUMM, SIFT, and histogram-based methods. By exploring these model-algorithm pairings, the research aims to advance the overall performance and capabilities of action detection systems. This study addresses an important research gap by conducting comprehensive evaluations and comparisons, examining the strengths, weaknesses, and unique characteristics of each model-algorithm combination. The findings offer valuable insights into the suitability and practicality of specific configurations in real-world scenarios. Moreover, the research endeavors to provide researchers and practitioners with informed decision-making tools when selecting appropriate approaches for action detection and keyframe extraction tasks. By presenting these findings, this research contributes to the existing knowledge base, facilitating further advancements and innovations in the field of video analysis and summarization.

The system architecture as mentioned in Figure 1 encompasses several key stages to facilitate accurate action detection in the context of crime analysis. Initially, the system acquires video inputs from the UCF Crime dataset, which provides a diverse range of crime-related scenarios for analysis. This dataset selection ensures the system's compatibility with real-world crime detection applications and enhances the generalizability of the research findings. Next, the system employs keyframe extraction techniques to identify and extract representative frames from the input videos. These keyframes capture crucial temporal instances that are essential for action detection. Various keyframe extraction methods can be utilized, such as the VSUMM algorithm, which employs clustering techniques to group similar frames and selects representative frames based on their relevance to the video content. Alternatively, the SIFT algorithm utilizes the scale-invariant feature transform to detect and match distinctive features in each frame, selecting frames with the most distinctive features for further analysis. Another option is the histogram-based keyframe extraction algorithm, which selects keyframes based on significant changes in color distribution between consecutive frames. The extracted keyframes are then subjected to evaluation by multiple models within the system. These models, which can include state-of-the-art action detection frameworks such as SlowFast, UniformerV2, and TIN, leverage deep learning architectures and advanced feature extraction methods to capture and analyze the spatio-temporal characteristics of the keyframes. Each model generates confidence scores indicating the likelihood of different action classes being present in the video. These confidence scores serve as a measure of the model's prediction accuracy and provide valuable insights into the action detection process. To refine the analysis and enhance the system's output, the system selects the top 5 accurate action classes based on the highest confidence scores obtained from the evaluation models. This step aims to prioritize the most relevant and probable action classes associated with the analyzed video, enabling focused and efficient crime detection.

Overall, the proposed system architecture combines keyframe extraction and evaluation models to optimize action detection in crime analysis scenarios. By leveraging diverse keyframe extraction algorithms and state-of-the-art evaluation models, the system demonstrates the potential for enhanced accuracy and efficiency in detecting crime-related actions. The utilization of the UCF Crime dataset further enhances the system's applicability to real-world crime detection tasks, making it a valuable contribution to the field of video-based crime analysis.

IV. Experimental Setup

A single Tesla K80 GPU with 12GB of memory is utilized for both training and evaluating the performance of the models. The system employs a modular setup, allowing the selection

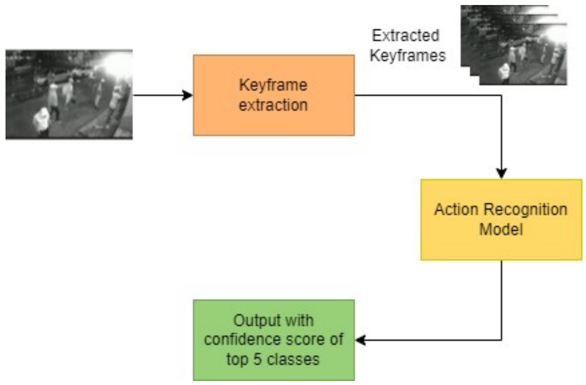

Fig. 1. Technical overview of the system to perform the comparative analysis of action detection models and keyframe extraction algorithms

of specific key frame extraction algorithms and models during training and inference. Notably, different models occupy varying amounts of GPU memory, necessitating adjustments to the batch size when switching between models. The training process involves training each model for 25 epochs on the UCF Crime dataset, with a focus on five selected classes to expedite training. Prior to model input, videos undergo pre-processing using the designated key frame extraction algorithm. After extensive experimentation, it is observed that a significant portion of the video's crucial frames can be captured by extracting only 10% of the total frames using the key frame extraction algorithm.

The Slow Fast model exhibits the highest memory footprint, consuming 150MB when loaded into memory, possibly due to its Siamese architecture. Consequently, the batch size for Slow Fast models is limited to 10, resulting in a training rate of approximately 1-1.5 hours per epoch. The accuracy curve, illustrated in section 1, demonstrates that the model initially reaches a local minima but further training drives it towards a global minima.The transformer-based UniformerV2 model demands the longest training time, primarily attributed to the intensive computations involved in the attention matrix for each frame. With a batch size of 20 and a learning rate of 5e-5, the UniformerV2 model requires approximately 1.5-1.75 hours per epoch. To comprehensively assess performance, all models are trained using every key frame extraction algorithm. The evaluation is conducted on five distinct classes: Explosion, Assault, Shooting, Fighting, and Road Accidents. Results are systematically tabulated in table II, capturing performance gains or losses for each algorithm-model combination. Additionally, the best results per class and their corresponding algorithm-model combinations are recorded for further analysis and comparison.

V. Results and Discussion

We evaluate the results of our experiments based on the training and inference time and at the same time accuracy of the model in predicting the appropriate crime class.

A. Training and Inference Time

Keyframe extraction [14] [15] is an integral part of anomaly detection as it directly affects the inference time and helps in selecting only those frames which have some anomaly and normal frames for the model to detect which class it belongs to. The training limit has been set to 25 epochs where each detection model has been paired with a different keyframe extraction algorithm.

TABLE I
Keyframe Extraction Algorithm and Model Vs Model Inference time

Detection Model	Keyframe extraction Algorithm	Time/ epoch (with keyframe extraction) (in hours)	Time/ epoch (w/o keyframe extraction) (in hours)	Percentage decrease in time
Slowfast	Image Histogram	∼1	∼2.88	∼65.28%
Slowfast	VSUMM algorithm	∼0.9	∼2.88	∼67.12%
Slowfast	SIFT algorithm	∼1.3	∼2.88	54.86%
UniformerV2	Image Histogram	∼1.4	∼3.84	63.54%
UniformerV2	VSUMM algorithm	∼1.2	∼3.84	68.75%
UniformerV2	SIFT algorithm	∼1.3	∼3.84	66.15%
TIN	Image Histogram	∼3	∼3.15	4.76 %
TIN	VSUMM algorithm	∼2.9	∼3.15	7.94%
TIN	SIFT algorithm	∼2.6	∼3.15	17.46%

From the table I, we can see that Slowfast and UniformerV2 detection models have the highest percentage decrease in time with 67.12% and 68.75% respectively with the keyframe extraction algorithm, providing a significant boost in inference time as compared to the TIN detection algorithm which has a very marginal boost time with keyframe extraction algorithms.

Keyframe extraction algorithms perform well with Slowfast due to their reliance on detecting motion and texture changes, aligning with Slowfast's emphasis on these features. This synergy reduces inference time. In contrast, the UniformerV2 Model, utilizing a self-attention mechanism, cross-checks every frame with all others. Keyframe extraction mitigates this by reducing the number of frames checked, optimizing inference time.

B. Model Accuracy

The VSUMM algorithm [13] excels in generating diverse and representative keyframe sets. Primarily used for video summarization, it efficiently compresses lengthy videos into concise summaries, capturing essential content aspects. VSUMM identifies keyframes representing various scenes or events, ensuring a compact summary that preserves visual content and context. Its versatility extends to video indexing, supplying representative frames for labeling and categorization.

The SIFT [16] algorithm is ideal for applications that require the detection and tracking of specific objects or features. For example, it can be used for object recognition, where the goal is to identify and track specific objects or patterns in the video. The SIFT algorithm can detect and match distinctive features, such as edges, blobs, and corners, across frames and use them to track the objects or patterns of interest. Additionally, SIFT can be used in surveillance applications to track the movement of people or vehicles and detect any anomalous behavior or activity.

The Histogram-based approach is ideal for applications that require keyframe selection based on color or texture similarity. For example, it can be used for image retrieval or classification, where the goal is to search for images or videos that are similar in color or texture to a given query. The Histogram-based approach can compute histograms of color or texture

TABLE II
KEYFRAME ALGORITHM AND MODEL VS AVG ACCURACY FOR ALL CLASSES

Model-Keyframe	Avg Accuracy for all classes (in %)
SlowFast without keyframes	79.20
SlowFast with Histogram	84.53
SlowFast with VSUMM	83.05
SlowFast with SIFT	82.10
UniformerV2 without keyframes	79.46
UniformerV2 with Histogram	80.14
UniformerV2 with VSUMM	81.73
UniformerV2 with SIFT	79.09
TIN without keyframes	78.76
TIN with Histogram	82.21
TIN with VSUMM	80.92
TIN with SIFT	83.60

Model-Keyframe	Accuracy for Explosion (in %)
SlowFast without keyframes	78.95
SlowFast with Histogram	81.65
SlowFast with VSUMM	85.43
SlowFast with SIFT	80.32
UniformerV2 without keyframes	79.54
UniformerV2 with Histogram	82.47
UniformerV2 with VSUMM	84.62
UniformerV2 with SIFT	79.57
TIN without keyframes	80.19
TIN with Histogram	82.57
TIN with VSUMM	83.90
TIN with SIFT	80.26

TABLE III
KEYFRAME ALGORITHM AND MODEL VS ACCURACY FOR EXPLOSION

features for each frame and select frames that are dissimilar to their neighbors based on the similarity of their histograms. Additionally, Histogram-based approaches can be used in video editing applications to select frames that match a specific color or style or to remove frames that are too similar or redundant.

The following observations can be inferred from Table II which describes multiple model accuracies when trained with different keyframe extraction algorithms.

VSUMM is well-suited for Explosion (Table III) and Shooting (Table IV) classes. Designed to offer a diverse and representative set of frames, it excels in capturing essential aspects of video content. In explosion videos, VSUMM effectively identifies crucial events like the initial blast, aftermath, and secondary explosions. Similarly, for shooting videos, the algorithm

Model-Keyframe	Accuracy for Shooting (in %)
SlowFast without keyframes	79.22
SlowFast with Histogram	80.25
SlowFast with VSUMM	85.21
SlowFast with SIFT	80.27
UniformerV2 without keyframes	81.90
UniformerV2 with Histogram	79.92
UniformerV2 with VSUMM	84.71
UniformerV2 with SIFT	80.32
TIN without keyframes	79.42
TIN with Histogram	81.34
TIN with VSUMM	84.63
TIN with SIFT	78.96

TABLE IV
KEYFRAME ALGORITHM AND MODEL VS ACCURACY FOR SHOOTING

Model-Keyframe	Accuracy for Fighting (in %)
SlowFast without keyframes	77.89
SlowFast with Histogram	82.92
SlowFast with VSUMM	81.36
SlowFast with SIFT	83.72
UniformerV2 without keyframes	76.98
UniformerV2 with Histogram	79.13
UniformerV2 with VSUMM	80.74
UniformerV2 with SIFT	81.53
TIN without keyframes	77.25
TIN with Histogram	83.94
TIN with VSUMM	81.81
TIN with SIFT	84.00

TABLE V
KEYFRAME ALGORITHM AND MODEL VS ACCURACY FOR FIGHTING

Model-Keyframe	Accuracy for Assault (in %)
SlowFast without keyframes	78.12
SlowFast with Histogram	81.39
SlowFast with VSUMM	80.51
SlowFast with SIFT	83.36
UniformerV2 without keyframes	77.59
UniformerV2 with Histogram	80.47
UniformerV2 with VSUMM	81.14
UniformerV2 with SIFT	84.83
TIN without keyframes	79.73
TIN with Histogram	83.14
TIN with VSUMM	82.57
TIN with SIFT	84.69

TABLE VI
KEYFRAME ALGORITHM AND MODEL VS ACCURACY FOR ASSAULT

Model-Keyframe	Accuracy for Road Accident (in %)
SlowFast without keyframes	77.23
SlowFast with Histogram	84.31
SlowFast with VSUMM	80.72
SlowFast with SIFT	81.89
UniformerV2 without keyframes	76.61
UniformerV2 with Histogram	80.43
UniformerV2 with VSUMM	81.62
UniformerV2 with SIFT	82.98
TIN without keyframes	77.01
TIN with Histogram	85.07
TIN with VSUMM	82.08
TIN with SIFT	83.45

TABLE VII
KEYFRAME ALGORITHM AND MODEL VS ACCURACY FOR ROAD ACCIDENT

Model-Keyframe	Accuracy for Normal (in %)
SlowFast without keyframes	77.47
SlowFast with Histogram	82.31
SlowFast with VSUMM	79.89
SlowFast with SIFT	83.58
UniformerV2 without keyframes	77.60
UniformerV2 with Histogram	83.81
UniformerV2 with VSUMM	80.12
UniformerV2 with SIFT	82.77
TIN without keyframes	77.34
TIN with Histogram	84.25
TIN with VSUMM	81.21
TIN with SIFT	83.14

TABLE VIII
KEYFRAME ALGORITHM AND MODEL VS ACCURACY FOR NORMAL

selects keyframes representing various events such as initial shots, reactions of people, and emergency response activities. VSUMM ensures a compact summary that encapsulates the overall sequence of events in these scenarios.

The SIFT algorithm may be better suited for Fighting(Table V) and Assault (Table VI) videos . This is because the SIFT algorithm is designed to detect and track specific objects or features in a video. In a fighting video, the main objects of interest are likely to be the people involved in the fight. The SIFT algorithm can detect and match distinctive features, such as facial features or clothing, across frames and use them to track the movements of the people involved in the fight. Similarly, the SIFT algorithm can help to identify any objects or weapons used in the case of assault, which can provide important evidence for law enforcement agencies.

For road accident videos, as indicated in Table VII, the Histogram-based approach is well-suited. Tailored for selecting frames based on color or texture similarity, it aligns with the crucial aspects of road accident scenarios—such as damaged vehicles, injuries, and traffic flow. By computing histograms of color or texture features for each frame and selecting dissimilar frames based on histogram similarity, this approach effectively identifies keyframes that capture essential information in accident videos.

For normal event videos as inferred from Table VIII, any of the keyframe extraction algorithms could be used, depending on the specific goals of the analysis. If the goal is to provide a representative set of frames for indexing or summarization, the VSUMM algorithm could be used. If the goal is to track specific objects or features, the SIFT algorithm could be used. If the goal is to select frames based on color or texture similarity, the Histogram-based approach could be used. A summary of accuracy vs epochs for different algorithms has been shown in 2

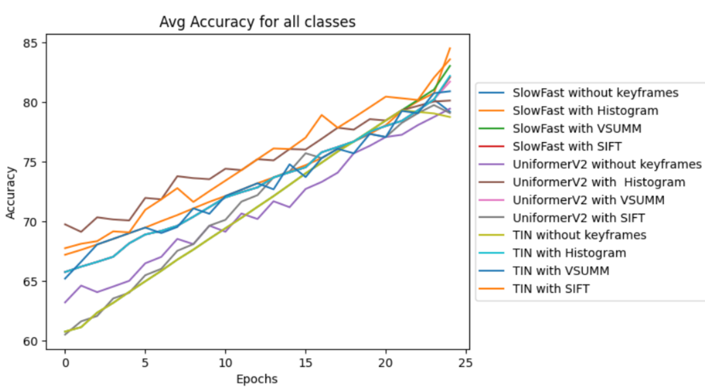

Fig. 2. Avg Accuracy vs Epochs

VI. Conclusion

Crime Detection is a promising field for the application of modern deep learning models to solve real world problems. The main issue with processing CCTV footages is the length of the videos; current Deep Learning models require all frames of a CCTV footage to analyse and detect the type of crime in the given video. However, we note that in most cases, 80% of any CCTV footage is normal activity and the model needs to capture the actions performed in the remaining 20% of the video.

We propose the use of key frame extraction algorithms to solve this issue and evaluate the performance of every algorithm on 3 different vision models. We conclude that the combination of using SlowFast model with the histogram based detection model is the overall most accurate and efficient method for detection of crime. We also evaluate the gains in inference time when working with key frame extraction algorithms and tabulate it in the above tables. Our work can act as an avenue for further research on pre-processing of long form videos for better performance on existing vision models.

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